Notes: Guvenen (Econometrica, 2009)

A Parsimonious Macroeconomic Model for Asset Pricing

Contents

1 Big picture

- **Question.** Can a parsimonious macroeconomic model explain major asset-pricing facts–a high equity premium, low and smooth risk-free rates, predictable returns, and countercyclical risk premia–without relying on extreme risk aversion or habit formation?
- **Core model ingredients.** The model adds two features to a real business cycle production economy: limited stock market participation and heterogeneity in the elasticity of intertemporal substitution (EIS).
- **Two household types.** Stockholders are a small group, calibrated to 20% of the population, who can hold equity and risk-free bonds. Non-stockholders are the majority and can only use the risk-free bond.
- **Preference heterogeneity.** Stockholders have a higher EIS, calibrated to 0.3. Non-stockholders have a very low EIS, calibrated to 0.1. Epstein–Zin preferences separate EIS from risk aversion.
- **Main mechanism.** Non-stockholders use the bond market to smooth aggregate labor-income risk. Since aggregate risk cannot be diversified away, stockholders end up absorbing that risk. This raises the volatility of stockholders' consumption growth and makes them demand a high equity premium.
- **Why wealth inequality matters.** The model is consistent with non-stockholders holding little aggregate wealth. Unlike earlier limited-participation models, the mechanism does not require large average interest payments to non-stockholders. It relies on the *cyclical timing* of bond-market payments.
- **Unconditional asset-pricing success.** In the inelastic-labor benchmark, the model generates an equity premium of 5.46%, a Sharpe ratio of 0.25, and a risk-free rate mean of 1.31%.
- **Interest-rate result.** The risk-free rate is relatively smooth because stockholders, with higher EIS and access to equity, are on the other side of the bond market. Their bond-supply curve is flatter than in representative-agent low-EIS models.
- **Asset-price dynamics.** The model produces a procyclical price-dividend ratio, mean reversion and long-horizon predictability in stock returns, and countercyclical expected excess returns, volatility, and Sharpe ratio.
- **Labor-supply result.** Cobb–Douglas leisure preferences damage asset-pricing performance and imply unrealistically smooth labor hours. GHH preferences preserve much of the asset-pricing performance and improve macro quantities, though not enough to solve all business-cycle shortcomings.
- **Economic takeaway.** Cross-sectional heterogeneity in market access and EIS can act like an endogenous insurance mechanism: the majority smooths consumption through bonds, while the minority stockholders bear concentrated aggregate risk and earn the equity premium.

2 Data, calibration, and measurement

2.1 What kind of paper this is

- The paper is a quantitative macroeconomic asset-pricing model rather than a reduced-form empirical paper.
- The main evidence comes from calibrating a two-agent production economy and comparing simulated moments to U.S. asset-pricing and macroeconomic data.
- The model uses the real business cycle framework as the baseline because standard RBC models have weak asset-pricing implications.

Interpretation. The paper's empirical discipline is moment matching and out-of-sample comparison to asset-price dynamics, not a causal regression design.

2.2 Key asset-pricing targets

The model is judged against several classic facts:

- a high average equity premium;
- a plausible volatility of stock returns and excess returns;
- a low and relatively smooth risk-free rate;
- a realistic price-dividend ratio and price-dividend volatility;
- mean reversion and return predictability;
- countercyclical expected returns, risk, and Sharpe ratios.

Interpretation. The key success is not merely matching the average equity premium. The paper also asks whether the model gets the *dynamics* of asset prices right.

2.3 Household groups and participation

- The population is normalized to one.
- A fraction μ are stockholders; the baseline value is $\mu = 0.2$.
- Stockholders can trade both stocks and one-period risk-free bonds.
- Non-stockholders cannot trade stocks but can trade the risk-free bond.
- The 20% participation rate reflects the pre-1990s period, before the large rise in U.S. household stock market participation.

Interpretation. The participation rate makes stockholders a small risk-bearing group. Aggregate risk is therefore concentrated among them when non-stockholders smooth labor-income risk through the bond market.

2.4 Preference parameters

- The stockholder EIS is set to $1/\rho^h = 0.3$.
- The non-stockholder EIS is set to $1/\rho^n = 0.1$.
- Both types have baseline risk aversion $\alpha^h = \alpha^n = 6$.
- Preference heterogeneity is mostly EIS heterogeneity, not risk-aversion heterogeneity.

Interpretation. A low EIS means a strong desire for smooth consumption over time. The majority non-stockholders therefore use the bond market aggressively for smoothing even though they have little wealth.

2.5 Technology and production calibration

- The aggregate firm uses Cobb–Douglas production with capital share $\theta = 0.30$.
- The technology shock is persistent, with quarterly persistence about 0.95.
- Capital adjustment costs are governed by a curvature parameter $\xi = 0.40$ in the baseline.
- Quarterly depreciation is 2%.
- Corporate leverage is calibrated to 15% of average capital.

Interpretation. Adjustment costs are crucial because they make stock prices volatile and procyclical, but they also create a trade-off with macroeconomic quantities, especially investment and consumption volatility.

2.6 Three labor-supply specifications

The paper studies three versions of period utility:

- **CONS:** inelastic labor supply and power utility over consumption;
- **CD:** Cobb–Douglas utility over consumption and leisure;
- **GHH:** Greenwood–Hercowitz–Huffman preferences, which remove wealth effects on labor supply and allow more flexible calibration of labor elasticity.

Interpretation. CONS is the cleanest version for understanding the asset-pricing mechanism. CD and GHH ask whether the mechanism survives in a full macro model with labor choice.

2.7 Simulation and solution method

- The model period is one month to approximate frequent asset trading.
- Simulated financial moments are reported annually; macro quantities are reported quarterly.
- Because markets are incomplete, allocations and prices must be solved jointly.
- The state vector includes capital, bond holdings of non-stockholders, and technology.
- The paper approximates equilibrium functions over the state space with multidimensional cubic splines rather than relying on local log-linearization.

Interpretation. The solution method is important because asset prices are very sensitive to approximation error, especially when state variables are volatile.

3 Models and specifications

3.1 Recursive preferences

For household type $i \in \{h, n\}$, utility is Epstein–Zin:

$$U_t^i = \left[(1-\beta)u^i(c_t, 1-l_t)^{1-\rho^i} + \beta\left(E_t[(U_{t+1}^i)^{1-\alpha^i}]\right)^{\frac{1-\rho^i}{1-\alpha^i}}\right]^{\frac{1}{1-\rho^i}}. \tag{1}$$

- α^i controls risk aversion.
- $1/\rho^i$ controls the EIS.
- c_t is consumption and l_t is labor supply.

Interpretation. Epstein–Zin preferences allow the paper to vary EIS without mechanically changing risk aversion. This is essential for showing that EIS heterogeneity, not risk-aversion heterogeneity, drives the results.

3.2 Production and technology

The aggregate firm produces

$$Y_t = Z_t K_t^\theta L_t^{1-\theta}, \tag{2}$$

with technology

$$\log Z_{t+1} = \phi \log Z_t + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2). \tag{3}$$

Interpretation. This is the standard RBC production block. The novelty is not production; it is who can hold equity and how households differ in intertemporal substitution.

3.3 Firm value and capital accumulation

The firm maximizes the discounted value of dividends:

$$P_t^s = \max_{\{I_{t+j}, L_{t+j}\}} E_t \sum_{j=1}^{\infty} \beta^j \frac{\Lambda_{t+j}}{\Lambda_t} D_{t+j}, \tag{4}$$

subject to adjustment-cost capital accumulation:

$$K_{t+1} = (1-\delta)K_t + \Phi\left(\frac{I_t}{K_t}\right)K_t. \tag{5}$$

Dividends are

$$D_t = Z_t K_t^\theta L_t^{1-\theta} - W_t L_t - I_t - (1-P_t^f)\chi\bar{K}. \tag{6}$$

Interpretation. The stock price is the value of the dividend stream, discounted by the marginal utility of stockholders. Adjustment costs make capital slow to adjust, which helps create volatile stock prices and procyclical price-dividend ratios.

3.4 Stockholder problem

A stockholder chooses consumption, labor, bond holdings, and stock holdings:

$$V^h(\omega^h; Y) = \max_{c,l,b',s'} \left[(1-\beta)u(c, 1-l)^{1-\rho^h} + \beta\left(E[V^h(\omega'; Y')^{1-\alpha^h}|Y]\right)^{\frac{1-\rho^h}{1-\alpha^h}}\right]^{\frac{1}{1-\rho^h}}, \tag{7}$$

subject to

$$c + P^f(Y)b' + P^s(Y)s' \le \omega + W(K, Z)l, \tag{8}$$

$$\omega' = b' + s'\left(P^s(Y') + D(Y')\right). \tag{9}$$

Interpretation. Stockholders are the only equity holders. Their stochastic discount factor prices the firm's dividend stream.

3.5 Non-stockholder problem

The non-stockholder solves the same problem except that

$$s' = 0. \tag{10}$$

Interpretation. Non-stockholders cannot use equity to smooth shocks. Their only intertemporal smoothing tool is the risk-free bond, so their low EIS makes bond demand steep and important.

3.6 Returns

The stock return, risk-free rate, and equity premium are

$$R^s_{t+1} = \frac{P^s_{t+1} + D_{t+1}}{P^s_t} - 1, \tag{11}$$

$$R^f_t = \frac{1}{P^f_t} - 1, \tag{12}$$

$$R^{ep}_{t+1} = R^s_{t+1} - R^f_t. \tag{13}$$

Interpretation. Asset-pricing success means the model can reproduce both the average and time-varying behavior of R^{ep} and R^f.

3.7 Approximate risk-free-rate intuition

The paper uses the approximation

$$r^f_t \approx -\log\beta + \rho^h E_t(\Delta \log c^h_{t+1}) + \kappa, \tag{14}$$

where κ collects volatility-related terms.

Interpretation. Epstein–Zin preferences help the model avoid an excessively high risk-free rate because the coefficient on expected consumption growth is tied to EIS rather than risk aversion.

3.8 Consumption-volatility decomposition

Stockholder consumption can be decomposed into a wage/capital-income component A^h_t and a bond-market transfer component a^h_t:

$$c^h_t = A^h_t + a^h_t. \tag{15}$$

The paper approximates stockholder consumption-growth variance as

$$\operatorname{var}(\Delta \log c^h) \approx \operatorname{var}(\Delta \log A^h) + \operatorname{var}\left(\Delta \frac{a^h}{A^h}\right) + 2\operatorname{cov}\left(\Delta \log A^h, \Delta \frac{a^h}{A^h}\right). \tag{16}$$

Interpretation. Average transfers are small, but their timing is powerful. When non-stockholders receive more insurance in recessions, stockholders' consumption becomes more procyclical and more volatile.

3.9 Conditional Sharpe ratio

The conditional Sharpe ratio can be understood through

$$SR_t \approx \alpha^h \times \sigma_t(\Delta \log c^h_{t+1}) \times \operatorname{corr}_t(\Delta \log c^h_{t+1}, R^{ep}_{t+1}). \tag{17}$$

Interpretation. The countercyclicality of the Sharpe ratio comes mainly from the countercyclicality of stockholders' conditional consumption-growth volatility.

3.10 Labor-supply utility specifications

The three period-utility specifications are:

$$\text{CONS:} \quad u(c, 1-l) = c^{1-\rho}, \tag{18}$$

$$\text{CD:} \quad u(c, 1-l) = \left(c^{\gamma}(1-l)^{1-\gamma}\right)^{1-\rho}, \tag{19}$$

$$\text{GHH:} \quad u(c, 1-l) = \left(c - \psi \frac{l^{1+\gamma}}{1+\gamma}\right)^{1-\rho}. \tag{20}$$

Interpretation. GHH preferences are useful because the Frisch labor-supply elasticity can be calibrated separately from the EIS. This matters because EIS heterogeneity is central to the paper.

4 Quantitative design and credibility

4.1 What disciplines the model

- Preference and technology parameters are chosen to match standard macro and finance targets.
- The shock volatility and risk aversion are calibrated so the model matches output volatility and a target annual Sharpe ratio of 0.25.
- The model is then evaluated on many additional moments: risk-free rate volatility, price-dividend dynamics, predictability, wealth distribution, and business-cycle quantities.

Interpretation. A model with enough flexibility can always match a few moments. The contribution is that two simple features–limited participation and EIS heterogeneity–explain a broad pattern of moments.

4.2 Why the EIS experiments are important

- Setting both groups' EIS to 0.3 lowers the equity premium from 5.46% to 2.44% and the Sharpe ratio from 0.25 to 0.16.
- Setting both groups' EIS to 0.1 raises volatility but does not raise the Sharpe ratio very much.
- Doubling non-stockholders' risk aversion from 6 to 12 has little effect.

Interpretation. These experiments isolate the key mechanism. Low EIS among non-stockholders matters much more than high risk aversion among non-stockholders.

4.3 Why the bond market matters

- Non-stockholders have low EIS, so their bond demand is steep.
- Stockholders have higher EIS and can also adjust equity/capital holdings, so their bond supply is flatter.
- Shocks to non-stockholder saving demand therefore lead to trade in the bond market rather than huge interest-rate volatility.

Interpretation. This is why the model can have both a low-EIS majority and relatively smooth interest rates, which is hard in representative-agent models.

4.4 Why the countercyclical premium emerges

- Non-stockholders own very little wealth.
- In recessions their wealth falls further, making their value function more curved and their smoothing demand stronger.
- Stockholders provide more insurance through the bond market and absorb more risk in recessions.
- Expected excess returns and Sharpe ratios therefore rise in recessions.

Interpretation. The countercyclical premium is not imposed directly by preferences. It comes from the interaction of incomplete markets, wealth inequality, and low-EIS non-stockholders.

4.5 Why labor supply is a stress test

- Endogenous labor supply can let households smooth marginal utility by changing hours.
- This can reduce the equity premium in production models.
- Cobb–Douglas preferences make the model perform worse.
- GHH preferences preserve the asset-pricing mechanism better and generate more realistic hours, but the macro fit remains imperfect.

Interpretation. The model is not a complete solution to macro-finance. Its strongest contribution is the asset-pricing and wealth-distribution mechanism.

4.6 What the paper cannot fully solve

- The model abstracts from long-run growth to preserve a stationary wealth distribution.
- Consumption remains too volatile and investment too smooth relative to the data in the best labor-supply versions.
- Stock market participation is taken as exogenous, not derived from fixed participation costs or life-cycle decisions.
- The model is calibrated rather than formally estimated.

Interpretation. The paper is best read as showing that limited participation plus EIS heterogeneity is a powerful mechanism, not as the final word on macro asset pricing.

5 Tables and figures

The figures and tables below are directly extracted from the paper and grouped to save space and improve comparison.

5.1 Table I + Table II: baseline parameters and inelastic-labor asset returns

Read: Table I:

- Stockholders have EIS 0.3 and non-stockholders have EIS 0.1.
- The stock-market participation rate is 20%.
- Risk aversion is 6 for both groups in the baseline.
- The adjustment-cost coefficient is 0.40 and leverage is 0.15.

Read: Table II:

- The CONS model produces an equity premium of 5.46%, close to the U.S. value of 6.17%.
- The model targets a Sharpe ratio of 0.25, compared with 0.32 in U.S. data.
- The model's average risk-free rate is 1.31%, close to the U.S. value of 1.94%.
- Equalizing EIS across groups sharply weakens the mechanism: with EIS 0.3/0.3, the equity premium falls to 2.44% and the Sharpe ratio to 0.16.
- Doubling non-stockholders' risk aversion has almost no effect on the main statistics.

Economic message. The key calibration is not extreme risk aversion. It is low EIS for the majority non-stockholders combined with stock-market exclusion.

TABLE I

BASELINE PARAMETRIZATION[a]

Parameter		Value
Calibrated Outside the Model		
β^*	Time discount rate	0.99
$1/\rho^h$	EIS of stockholders	0.3
$1/\rho^n$	EIS of non-stockholders	0.1
μ	Participation rate	0.2
ϕ^*	Persistence of aggregate shock	0.95
θ	Capital share	0.30
ξ	Adjustment cost coefficient	0.40
δ^*	Depreciation rate	0.02
$\underline{B}$	Borrowing limit	$6\overline{W}$
χ	Leverage ratio	0.15
Calibrated Inside the Model (to Match Targets)		
σ_ε^*	Standard deviation of shock (%)	1.5/1.5/1.1
$\alpha^h = \alpha^n$	Relative risk aversion	6

[a]The asterisks indicate that the reported value refers to the implied quarterly value for a parameter that is cali

Table 1: Baseline parametrization

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TABLE II

UNCONDITIONAL MOMENTS OF ASSET RETURNS: MODEL WITH INELASTIC LABOR SUPPLY

	U.S. Data	CONS Model			
α^h, α^n		6/6	6/6	6/6	6/**12**
$1/\rho^h, 1/\rho^n$		0.3/0.1	0.3/**0.3**	**0.1**/0.1	0.3/0.1
A. Stock and Bond Returns					
$E(R^{ep})$	6.17 (1.99)	5.46	2.44	7.65	5.52
$\sigma(R^{ep})$	19.4 (1.41)	21.9	15.3	27.2	22.0
$\sigma(R^s)$	19.3 (1.38)	20.6	14.7	27.0	20.8
$E(R^{ep})/\sigma(R^{ep})$	0.32 (0.11)	0.25[a]	0.16	0.28	0.25
$E(R^f)$	1.94 (0.54)	1.31	3.20	0.24	1.35
$\sigma(R^f)$	5.44 (0.62)	6.65	4.55	8.52	6.71
B. Price–Dividend Ratio					
$E(P^s/D)$	22.1 (0.58)	27.2	25.9	29.5	27.1
$\sigma(\log(P^s/D))$	26.3 (1.67)	26.6	13.8	38.7	26.9
$\sigma(\Delta \log D)$	13.4 (0.94)	19.1	14.0	24.2	19.1
C. Consumption Growth Volatility					
$\frac{\sigma(\Delta \log c^h)}{\sigma(\Delta \log c^n)}$	>1.5–2.0[b]	2.42	0.78	1.12	2.44

[a]The Sharpe ratio of 0.25 is one of the two empirical targets in the calibration. All statistics are reported in annualized percentages. Annual returns are calculated by summing log monthly returns. The numbers in parentheses in the second column are the standard errors of the statistics to reflect the sampling variability in the U.S. data.

[b]The value reported here represents an approximate lower bound for this ratio based on the empirical evidence discussed in Section 5.1.

Table 2: Unconditional moments with inelastic labor supply

5.2 Table III + Figure 1: elastic labor supply and bond-market intuition

Read: Table III:

- Cobb–Douglas labor preferences reduce the equity premium to 2.65% and the Sharpe ratio to 0.17.
- GHH preferences perform better: the equity premium is 4.21%, the Sharpe ratio is 0.24, and risk-free-rate volatility falls to 4.10%.
- Lowering the Frisch elasticity in the GHH model modestly weakens asset-pricing performance.

Read: Figure 1:

- In a representative-agent low-EIS model, bond supply is vertical and bond demand shifts create large bond-price movements.
- In the limited-participation model, non-stockholder demand is steep, but stockholder supply is flatter because stockholders have higher EIS and another asset.
- Therefore, shocks to saving demand move quantities more and bond prices less.

Economic message. GHH preferences show that the limited-participation mechanism can survive endogenous labor supply. Figure 1 explains why the risk-free rate is smoother than in many low-EIS representative-agent models.

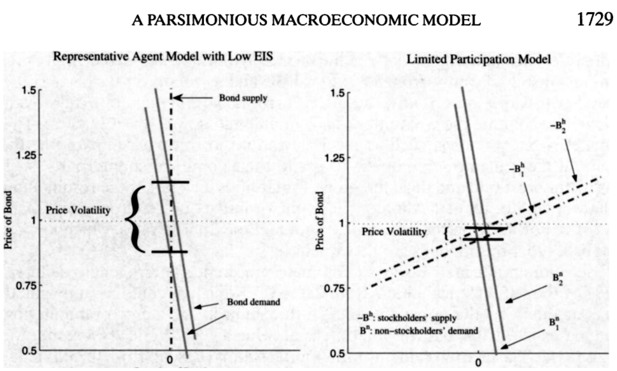

Figure 1: Determination of bond price volatility

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TABLE III

UNCONDITIONAL MOMENTS OF ASSET RETURNS: MODEL WITH ELASTIC LABOR SUPPLY

	U.S. Data	Model			BCF	D-D
Endogenous Leisure?		CD	$GHH_{Baseline}$	$GHH_{(Low\ Frisch)}$	Linear	None
A. Stock and Bond Returns						
$E(R^{ep})$	6.17 (1.99)	2.65	4.21	4.03	6.63	5.23
$\sigma(R^{ep})$	19.4 (1.41)	15.4	17.4	18.1	—	25.3
$\sigma(R^s)$	18.7 (1.38)	14.8	16.5	17.9	18.4	24.3
$E(R^{ep})/\sigma(R^{ep})$	0.32 (0.11)	0.17	0.24	0.22	0.36[a]	0.21
$E(R^f)$	1.94 (0.54)	2.87	1.42	1.73	1.20	1.32
$\sigma(R^f)$	5.44 (0.62)	4.91	4.10	4.46	24.6	10.61
B. Price–Dividend Ratio						
$E(P^s/D)$	22.1 (0.58)	25.7	24.7	25.9	—	—
$\sigma(\log(P^s/D))$	26.3 (1.67)	13.6	17.8	19.1	—	—
$\sigma(\Delta \log D)$	13.4 (0.94)	14.8	11.2	11.9	—	4.55

[a]BCF reported the time average of the conditional Sharpe ratio, $E_t(R^{ep})/\sigma_t(R^S)$, instead of the unconditional Sharpe ratio reported in the present paper. The statistics from BCF refer to their benchmark model (called the preferred two sector model), which has the best overall performance. The statistics from Danthine and Donaldson (2002, D-D) are from their Table 6, right column with smoothed dividends, which generates the best overall performance. A dash indicates that the corresponding statistic has not been reported in that paper. In the fifth column, the Frisch elasticity is set to 0.5 and the σ_E is recalibrated to 1.3% per quarter to match output volatility.

further falls to 4.1% with endogenous labor supply below). Although this figure is higher than the empirical values, the low volatility of the interest rate has

Table 3: Unconditional moments with elastic labor supply

5.3 Table IV + Table V: mean reversion and return predictability

Read: Table IV:

- U.S. excess returns exhibit mean reversion, though individual autocorrelations are noisy.
- The model generates negative autocorrelations and similar signs for aggregated autocorrelation measures.

Read: Table V:

- The price-dividend ratio predicts stock returns with negative coefficients.
- Predictability grows with horizon in both the U.S. data and the model.
- For stock returns, model R^2 reaches 0.35 at seven years.
- For excess returns, model predictability is qualitatively right but quantitatively weaker: seven-year R^2 is 0.12 versus 0.33 in the U.S. data.

Economic message. The model does not only match unconditional moments. It also produces the kind of long-horizon predictability associated with time-varying risk premia.

he finding of mean reversion is a clear departure from the martingale hy-
ıesis of returns and is closely linked to the predictability of returns doc-
ːnted by Campbell and Shiller (1988), among others. To show this, I first
ess log stock returns on the log price–dividend ratio using the historical

TABLE IV

THE AUTOCORRELATION STRUCTURE OF KEY FINANCIAL VARIABLES[a]

	Autocorrelation at Lag j (years)				
	1	2	3	5	7
$r^s - r^f$					
U.S. data	0.03	−0.22	0.08	−0.14	0.10
Model	−0.03	−0.03	−0.02	−0.02	−0.02
$\sum_{i=1}^{j} \rho[(r^s - r^f)_t, (r^s - r^f)_{t-i}]$					
U.S. data	0.03	−0.19	−0.11	−0.29	−0.15
Model	−0.03	−0.06	−0.08	−0.13	−0.15

[a]Statistics from the model are calculated from annualized values of each variable, except for the stock price, which is taken to be the value at the end of the year.

Table 4: Autocorrelation structure of key financial variables

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TABLE V

LONG-HORIZON REGRESSIONS ON PRICE–DIVIDEND RATIO[a]

	U.S. Data		Model	
Horizon (k)	Coefficient	R^2	Coefficient	R^2
A. Stock Returns				
1	−0.21	0.07	−0.19	0.09
2	−0.36	0.12	−0.40	0.16
3	−0.41	0.13	−0.49	0.22
5	−0.70	0.23	−0.77	0.28
7	−0.87	0.27	−0.91	0.35
B. Excess Returns				
1	−0.22	0.09	−0.11	0.02
2	−0.39	0.14	−0.20	0.04
3	−0.47	0.15	−0.29	0.06
5	−0.77	0.26	−0.42	0.10
7	−0.94	0.33	−0.49	0.12

[a]The table reports the coefficients and R^2 values of the regression of log

Table 5: Long-horizon regressions on the price-dividend ratio

5.4 Figure 2 + Table VI: countercyclical risk premia and the source of consumption volatility

Read: Figure 2:

- Expected excess returns are countercyclical, with correlation around −0.55 with output.
- The Sharpe ratio is also countercyclical, with correlation around −0.63.
- The model therefore generates higher compensation for risk in recessions.

Read: Table VI:

- Average interest-transfer components are tiny: about 1.1% of stockholder consumption and 0.5% of non-stockholder consumption.
- Yet the transfer component and its covariance account for most stockholder consumption-growth volatility.
- Stockholder consumption growth volatility is 3.61%, while non-stockholder volatility is 1.48%.

Economic message. The model's amplification is not about large average bond payments. It is about small payments arriving at exactly the states where they transfer aggregate risk from non-stockholders to stockholders.

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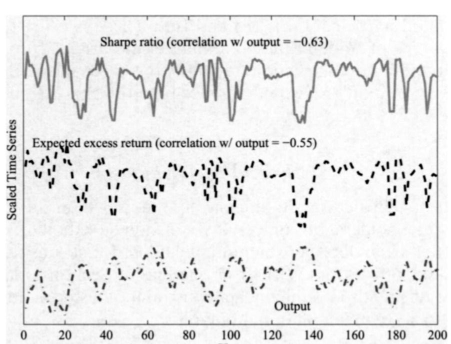

FIGURE 2.—The cyclical behavior of expected excess returns and the Sharpe ratio.

Figure 2: Cyclical behavior of expected returns and Sharpe ratio

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TABLE VI

DECOMPOSITION OF CONSUMPTION VOLATILITY AND EQUITY PREMIUM[a]

	Fraction of $E(c^i)$ Explained by	
	$E(A^i)/E(c^i)$	$E(a^i)/E(c^i)$
Stockholders	1.011	0.011
Non-stockholders	0.995	0.005

	Fraction of $\sigma^2(\Delta \log c^i)$ Accounted for by			
	$\sigma^2(\Delta \log c^i)$	$\sigma^2(\Delta \log A^i)$	$\sigma^2(\Delta \frac{a^i}{A^i})$	$2\sigma(\Delta \log A^i, -\Delta \frac{a^i}{A^i})$
Stockholders	$(3.61\%)^2$	0.185	0.340	0.475
Non-stockholders	$(1.48\%)^2$	3.13	0.61	−2.74

[a]$i = h, n$. See the text and equation (10) for the definitions of a^i and A^i.

.01% for both stockholders and non-stockholders. However, interest pay-

Table 6: Decomposition of consumption volatility and equity premium

5.5 Figure 3 + Figure 4: value-function curvature and recession risk

Read: Figure 3:

- Non-stockholders' value function is much steeper at low wealth than stockholders' value function.
- Because non-stockholders own little wealth, business-cycle movements change their marginal value of wealth more strongly.

Read: Figure 4:

- Around recessions, non-stockholders become more averse to risk.
- Stockholders' attitude toward risk is much more stable because their wealth share is large.

Economic message. Wealth inequality is not a side result. It is central for countercyclical risk premia: poor non-stockholders demand more insurance in recessions, and rich stockholders supply it.

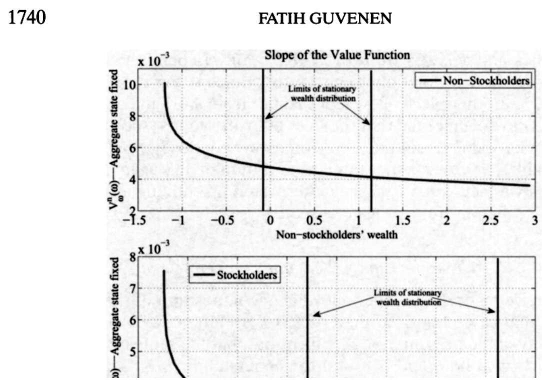

Figure 3: Change in the slope of value functions

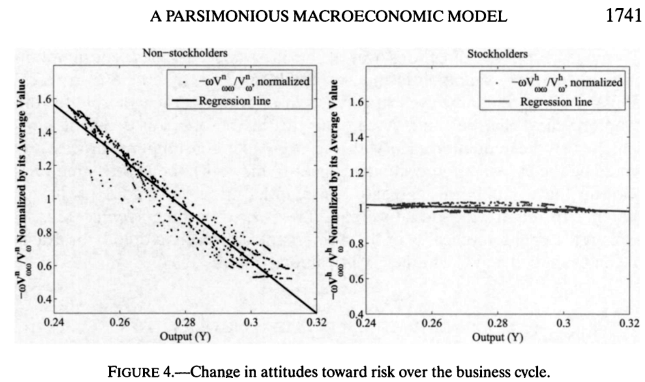

Figure 4: Change in attitudes toward risk over the business cycle

5.6 Table VII + Figure 5: business-cycle performance and model trade-offs

Read: Table VII:

- The CD model makes labor hours far too smooth: hours volatility relative to output is 0.07 versus 0.80 in the data.
- GHH baseline improves hours volatility to 0.50, closer to the data.
- Both CD and GHH still imply consumption volatility too high and investment volatility too low relative to the data.

Read: Figure 5:

- Raising the adjustment-cost parameter makes investment more volatile and consumption smoother.
- This improves macro quantities but lowers the equity premium and Sharpe ratio.
- Increasing stockholder risk aversion raises the equity premium without doing much for macro volatilities.

Economic message. The model faces a macro-finance trade-off. Lower adjustment costs help quantities but weaken asset pricing; higher risk aversion helps asset pricing but is less attractive economically.

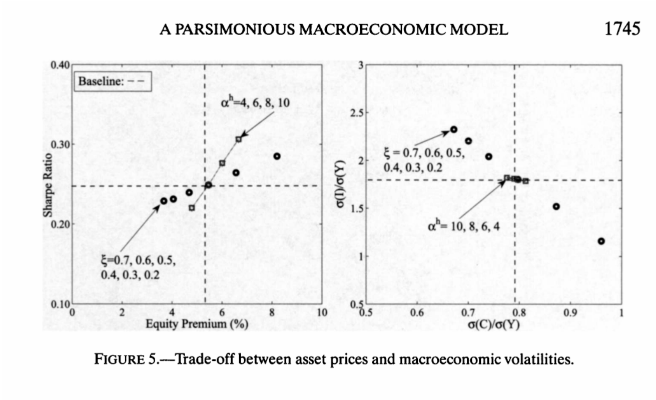

Figure 5: Trade-off between asset prices and macroeconomic volatilities

6.1. *Business-Cycle Implications*

Table VII displays standard business cycle statistics from the U.S. data as well as their counterparts from the CD and GHH models. As before, the last two columns report the corresponding figures from BCF and Danthine and Donaldson (2002) for comparison.

TABLE VII
BUSINESS CYCLE STATISTICS IN THE U.S. DATA AND IN THE MODEL[a]

	Data	Model			BCF	D-D
Leisure Preferences		CD	$\text{GHH}_{\text{Baseline}}$	$\text{GHH}_{\text{Low Frisch}}$	Linear	None
			Volatility			
$\sigma(Y)$	1.89 (0.21)	1.97	1.95	1.96	1.97	1.77
$\sigma(C)/\sigma(Y)$	0.40 (0.04)	0.92	0.78	0.75	0.69	0.82
$\sigma(I)/\sigma(Y)$	2.39 (0.06)	1.38	1.76	1.86	1.67	1.72
$\sigma(L)/\sigma(Y)$	0.80 (0.05)	0.07	0.50	0.32	0.51	—

Table 7: Business-cycle statistics in the U.S. data and the models

5.7 Figure 6: wealth distribution

Read:

- Stockholders hold roughly 92% of aggregate wealth in the GHH model.
- Non-stockholders hold roughly 8%.
- These numbers are broadly consistent with U.S. data, where stockholders hold most wealth and financial assets.
- This is a key difference from earlier limited-participation models that required non-stockholders to hold large wealth in order to generate large interest payments.

Economic message. The model's high equity premium is consistent with extreme wealth inequality. The mechanism uses cyclical bond-market insurance, not large average wealth holdings by non-stockholders.

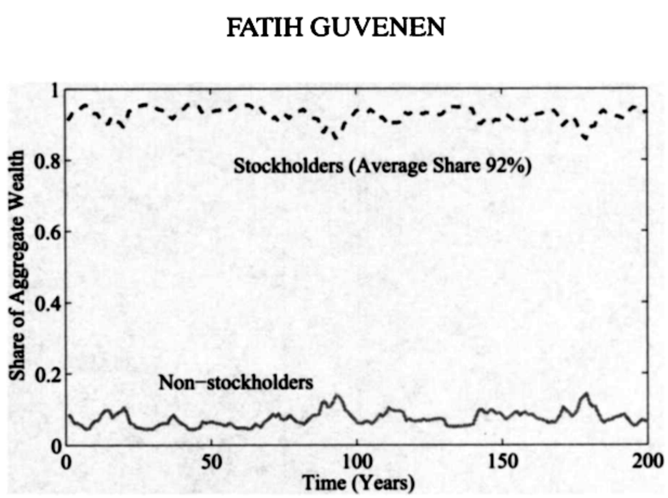

Figure 6: Evolution of wealth distribution over time

6 Short summary

- Guvenen studies a two-agent macro asset-pricing model with limited stock-market participation and heterogeneous EIS.
- Stockholders are 20% of the population and can hold equity and bonds.
- Non-stockholders are 80% of the population and can only use risk-free bonds.
- Non-stockholders have low EIS and therefore strong desire for consumption smoothing.
- The bond market reallocates aggregate labor-income risk from non-stockholders to stockholders.
- Because stockholders are a small group, the aggregate risk is concentrated in their consumption.
- This produces a high equity premium with moderate risk aversion.
- The inelastic-labor benchmark matches the equity premium and Sharpe ratio fairly well.
- The model also produces a relatively low and smooth risk-free rate.
- EIS heterogeneity is essential; risk-aversion heterogeneity is not.
- The model produces countercyclical expected returns, return volatility, and Sharpe ratios.
- The model generates long-horizon return predictability through a procyclical price-dividend ratio.
- The countercyclical equity premium comes from low-wealth non-stockholders becoming effectively more risk averse in recessions.
- GHH labor preferences preserve the asset-pricing mechanism better than Cobb–Douglas preferences.
- The model still struggles with some macro quantities: consumption is too volatile and investment is too smooth.
- The implied wealth distribution is highly skewed toward stockholders, matching an important fact in the data.
- The key lesson is that asset prices depend not only on aggregate consumption, but on who bears aggregate risk.

7 Conclusion

7.1 Core conclusion

Guvenen shows that limited stock-market participation combined with EIS heterogeneity can generate a high and countercyclical equity premium in a standard production economy. The model's mechanism is a bond-market insurance channel: low-EIS non-stockholders smooth aggregate labor-income risk through risk-free bonds, and stockholders absorb that risk.

7.2 Economic intuition

The intuition is that non-stockholders care strongly about smoothing consumption but cannot use equity. They therefore rely on bonds. Since aggregate risk cannot disappear, smoothing for the majority must come from someone else bearing the risk. Stockholders are the small group on the other side of the bond market. Their consumption becomes more volatile and more procyclical, so they require a large premium to hold equity.

7.3 Takeaway

The paper moves macro asset pricing away from representative-agent logic. Aggregate consumption may look smooth, but asset prices are determined by the marginal investors who hold risky assets. If those marginal investors are a small, wealthy group absorbing the economy's aggregate risk, then high and time-varying risk premia can arise from a parsimonious macro model.